



TERMS AND ABBREVIATIONS

The following terms and abbreviations are used in tables 1–19:

bls	below land surface
bmp	below the measuring point of the well casing
mi	mile
ft	feet
hhmm	24-hour time
mm/dd/yy	numerical date format for two-digit month/two-digit day/ and the last two digits of the year
C	Celsius
cm	centimeter
°	degree
in. or "	inch
μS/cm	microsiemens per centimeter at 25°Celsius
pH units	are the negative base-10 log of the hydrogen-ion activity in moles per liter
mg/L	milligrams per liter
μg/L	micrograms per liter
pCi/L	picocuries per liter
dissolved	refers to that fraction of material in a water sample that passes through a 0.45-μm membrane filter
>	greater than
<	less than
g/kg	grams per kilogram
mg/kg	milligrams per kilogram
pCi/g	picocuries per gram
μg/g	micrograms per gram
NAD 83	North American Datum 1983
NAVD 88	North American Vertical Datum 1988

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000

[$\mu\text{S}/\text{cm}$, microsiemens per centimeter at 25 degrees Celsius; mg/L , milligrams per liter; $\mu\text{g}/\text{L}$, micrograms per liter; $\text{p}^{\text{C}}\text{V}\text{L}$, picocuries per liter; $\leq$, less than; ND, no data available; E, value estimated by laboratory; -, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Well number	Date (mm/dd/yy)	Time (hh:mm)	Specific conductance, lab ($\mu\text{S}/\text{cm}$)	pH, laboratory (standard units)	Hardness total (mg/L as CaCO_3)	Calcium, dissolved (mg/L as Ca)	Magnesium, dissolved (mg/L as Mg)	Potassium, dissolved (mg/L as K)	Sodium adsorption ratio	Sodium, dissolved (mg/L as Na)	Sodium (percent)	Acid-neutralizing capacity, titration to pH 4.5, lab (mg/L as CaCO_3)	Chloride, dissolved (mg/L as Cl)
DTX1	01/10/00	1345	4,100	7.3	2,100	477	221	3.5	4	372	28	319	49.7
DTX1	04/14/00	1220	4,150	7.2	2,000	453	198	3.3	3	331	27	318	47.5
DTX1	07/07/00	0940	4,050	7.3	2,000	459	209	3.1	3	355	28	315	49.9
DTX1	10/17/00	1200	4,070	7.3	1,700	467	124	3.5	4	331	30	316	49.1
DTX2	01/10/00	1520	4,200	7.1	2,000	460	194	8.6	4	364	29	406	45.6
DTX2	04/14/00	1345	4,240	7.0	2,000	495	194	7.3	3	360	28	404	40.2
DTX2	07/07/00	1050	4,250	7.1	2,100	472	214	7.8	4	392	29	405	36.0
DTX2	10/17/00	1005	4,230	7.0	1,700	479	122	8.5	4	382	33	409	40.2
DTX3	01/10/00	1035	1,410	7.4	720	170	69.9	6.2	.9	58.2	15	274	17.1
DTX3	04/11/00	1015	1,880	7.3	990	240	94.2	7.3	1	72.8	14	314	26.0
DTX3	07/10/00	1030	2,020	7.4	1,100	266	102	7.3	1	94.1	16	328	28.6
DTX3	10/11/00	1030	2,100	7.1	1,100	263	102	7.5	1	99.5	17	338	28.7
DTX4	01/06/00	0955	2,950	7.0	1,700	531	86.9	8.8	2	160	17	353	15.2
DTX4	04/11/00	1145	3,140	6.9	1,800	551	90.8	7.2	2	158	16	424	16.1
DTX4	07/11/00	1010	3,070	6.9	1,700	526	89.1	7.5	2	169	18	346	17.4
DTX4	10/11/00	1215	3,200	6.9	1,800	563	95.4	8.8	2	179	18	348	16.1

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

[μS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; μg/L, micrograms per liter; pCi/L, picocuries per liter; <, less than; ND, no data available; E, value estimated by laboratory; ~, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Well number	Date (mm/dd/yy)	Time (hhmm)	Aluminum, dissolved (μg/L as Al)	Anti-mony, dissolved (μg/L as Sb)	Arsenic, dissolved (μg/L as As)	Barium, dissolved (μg/L as Ba)	Beryllium, dissolved (μg/L as Be)	Boron, dissolved (μg/L as B)	Bromide, dissolved (mg/L as Br)	Cadmium, dissolved (μg/L as Cd)	Chromium, dissolved (μg/L as Cr)	Cobalt, dissolved (μg/L as Co)	Copper, dissolved (μg/L as Cu)
DTX1	01/10/00	1345	7	<2	E1	9	<2	670	0.67	<2.0	<1.0	<2	8
DTX1	04/14/00	1220	<4	<1	4	8	<1	530	.71	<1.0	<.8	<1	14
DTX1	07/07/00	0940	<1	<1	E2	8	<1	550	.33	<1.0	<.8	1	9
DTX1	10/17/00	1200	2	<2	E2	8	<2	590	.70	<2.0	<.8	2	12
DTX2	01/10/00	1520	18	<2	<2	16	<2	310	.53	<2.0	<1.0	5	6
DTX2	04/14/00	1345	<3	<1	E2	15	<1	290	.40	<1.0	<.8	4	11
DTX2	07/07/00	1050	<1	<1	E1	14	<1	350	.41	<1.0	<.8	5	8
DTX2	10/17/00	1005	<2	<2	E1	14	<2	360	.48	<2.0	<.8	7	11
DTX3	01/10/00	1035	2	<1	<2	23	<1	210	.22	<1.0	E.6	<1	3
DTX3	04/11/00	1015	<6	<1	<2	23	<1	220	.28	<1.0	<.8	<1	3
DTX3	07/10/00	1030	<6	<1	<2	21	<1	230	.29	<1.0	<.8	<1	5
DTX3	10/11/00	1030	<1	<1	<2	21	<1	230	.27	<1	1.7	1	3
DTX4	01/06/00	0955	<2	<2	E1	14	<2	290	.22	<2.0	<.8	<1	5
DTX4	04/11/00	1145	<29	<2	E1	13	<2	270	.23	<2.0	<1.0	<2	6
DTX4	07/11/00	1010	<1	<2	<2	12	<2	240	.29	<2.0	<.8	<2	7
DTX4	10/11/00	1215	<2	E.1	<2	14	<1	310	.31	E.1	2.6	1	6

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

[pS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; <, less than; ND, no data available; E, value estimated by laboratory; -, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Well number	Date (mm/dd/yy)	Time (hh:mm)	Aluminum, dissolved (µg/L as Al)	Antimony, dissolved (µg/L as Sb)	Arsenic, dissolved (µg/L as As)	Barium, dissolved (µg/L as Ba)	Beryllium, dissolved (µg/L as Be)	Boron, dissolved (µg/L as B)	Bromide, dissolved (mg/L as Br)	Cadmium, dissolved (µg/L as Cd)	Chromium, dissolved (µg/L as Cr)	Cobalt, dissolved (µg/L as Co)	Copper, dissolved (µg/L as Cu)
DTX5	01/06/00	1310	2	<2	<2	16	<2	400	0.20	<2.0	0.9	3	6
DTX5	04/11/00	1345	<2	<2	<2	13	<2	330	.19	<2.0	<1.0	3	6
DTX5	07/11/00	1150	<1	<2	<2	13	<2	390	2.1	<2.0	<.8	2	7
DTX5	10/16/00	1015	<2	1	<2	14	<1	470	.25	<.1	<.8	2	7
DTX6	01/06/00	1530	3	<2	E1	9	<2	360	.15	<2.0	<.8	<2	7
DTX6	04/14/00	1000	<2	<1	2	9	<1	340	.17	<1.0	.8	<1	16
DTX6	07/13/00	1300	3	<1	<2	9	<1	340	.23	<1.0	<3.2	<1	9
DTX6	10/11/00	1400	3	<1	<2	8	<1	350	.17	E.1	3.0	1	9
DTX8A	01/13/00	1315	2	<1	<2	22	<1	260	.26	<1.0	E.8	<1	2
DTX8A	04/13/00	1415	<2	<1	<2	21	<1	240	.25	<1.0	<.8	<1	4
DTX8A	07/10/00	1505	<4	<1	<2	19	<1	260	.26	<1.0	<.8	<1	3
DTX8A	10/16/00	1435	3	E.03	<2	17	<1	260	.27	<.1	<.8	1	2
DTX10A	01/11/00	1130	2	<1	<2	16	<1	230	0.25	<1.0	1.5	<1	3
DTX10A	04/17/00	1415	<4	<2	<2	13	<2	240	.26	<2.0	<1.0	<2	6
DTX10A	07/13/00	1020	3	<1	<2	12	<1	250	.25	<1.0	<3.2	1	6
DTX10A	10/18/00	1040	<2	<1	<2	12	<1	250	.28	<.1	<.8	1	5

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

[uS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; <, less than; ND, no data available; E, value estimated by laboratory; --, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDL, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Well number	Date (mm/dd/yy)	Time (hh:mm)	Aluminum, dissolved (µg/L as Al)	Antimony, dissolved (µg/L as Sb)	Arsenic, dissolved (µg/L as As)	Barium, dissolved (µg/L as Ba)	Beryllium, dissolved (µg/L as Be)	Boron, dissolved (µg/L as B)	Bromide, dissolved (mg/L as Br)	Cadmium, dissolved (µg/L as Cd)	Chromium, dissolved (µg/L as Cr)	Cobalt, dissolved (µg/L as Co)	Copper, dissolved (µg/L as Cu)
D6	01/11/00	1355	121	<1	2	<1	<1	890	4.32	<1.0	<4.0	1	15
D6	04/13/00	0945	1	<1	1E9	6	<1	1E750	4.32	<1.0	<8	6	61
D6	07/11/00	1345	<1	<7	1E1	<7	<7	820	3.95	<7.0	<8	<7	35
D6	10/12/00	1300	<1	<1	3	5	<1	970	4.15	.1	<8	6	33
D13	01/12/00	1005	<1	<1	<2	20	<1	80	.18	<1.0	9	<1	3
D13	04/17/00	1115	<7	<1	<2	17	<1	80	.17	<1.0	<8	<1	2
D13	07/07/00	1230	<1	<1	<2	19	<1	100	.15	<1.0	1E5	<1	2
D13	10/16/00	1215	<1	<1	<2	24	<1	110	.19	1E02	<8	1	3
D17	01/07/00	1500	3	<1	1E1	61	<1	60	.08	<1.0	<8	<1	<1
D17	04/10/00	1415	<1	<1	1E1	58	<1	60	.09	<1.0	<8	<1	<1
D17	07/06/00	1320	<1	<1	1E1	56	<1	90	.09	<1.0	<8	<1	1
D17	10/10/00	1245	<1	<1	1E1	59	<1	60	.09	<1	<8	1	1
D25	01/12/00	1245	4	<1	3	20	<1	300	1.57	<1.0	2.6	4	9
D25	04/10/00	1615	1	<1	3	19	<1	250	1.26	<1.0	<1.0	3	11
D25	07/06/00	1500	<1	<1	3	17	<1	440	10.05	<1.0	<8	3	10
D25	10/10/00	1455	<2	<2	3	23	<2	400	1.38	<2.0	4.4	5	9

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

[uS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; <, less than; ND, no data available; E, value estimated by laboratory; -, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day.]

Well number	Date (mm/dd/yy)	Time (hhmm)	Aluminum, dissolved (µg/L as Al)	Antimony, dissolved (µg/L as Sb)	Arsenic, dissolved (µg/L as As)	Barium, dissolved (µg/L as Ba)	Beryllium, dissolved (µg/L as Be)	Boron, dissolved (µg/L as B)	Bromide, dissolved (mg/L as Br)	Cadmium, dissolved (µg/L as Cd)	Chromium, dissolved (µg/L as Cr)	Cobalt, dissolved (µg/L as Co)	Copper, dissolved (µg/L as Cu)
D29	01/07/00	1130	6	<2	<2	9	<2	180	0.18	<2.0	<1.0	<2	6
D29	04/10/00	1215	3	<1	E1	9	<1	190	.18	<1.0	<1.0	1	10
D29	07/06/00	1105	<1	<1	<2	9	<1	190	.20	<1.0	.8	2	8
D29	10/12/00	1110	<1	<1	<2	8	<1	200	.22	<1.0	<.8	1	7
D30	01/11/00	1600	2	<1	E1	5	<1	490	.72	<1.0	<1	1	5
D30	04/13/00	1200	<2	<1	3	10	<1	440	.70	<1.0	.9	2	18
D30	07/10/00	1245	<13	<1	<2	10	<1	450	.66	<1.0	<.8	2	12
D30	10/12/00	1450	1	<1	E2	10	<1	460	.64	<1.0	<.8	2	9

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

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Well number	Date (mm/dd/yy)	Time (hhmm)	Iron, dissolved (µg/L as Fe)	Lead, dissolved (µg/L as Pb)	Manganese, dissolved (µg/L as Mn)	Mercury, dissolved (µg/L as Hg)	Molybdenum, dissolved (µg/L as Mo)	Nickel, dissolved (µg/L as Ni)	Selenium, dissolved (µg/L as Se)	Silver, dissolved (µg/L as Ag)	Strontium, dissolved (µg/L as Sr)	Zinc, dissolved (µg/L as Zn)	Uranium, natural (µg/L as U)
DTX1	01/10/00	1345	<10	<2	90	<0.2	6	22	<2	<2	5,960	7	54
DTX1	04/14/00	1220	<30	<1	77	<2	6	4	6	<1	5,340	<14	54
DTX1	07/07/00	0940	<30	<1	96	<2	6	16	E2	<1	5,640	4	48
DTX1	10/17/00	1200	<30	<2	130	<2	7	31	3	<2	5,460	8	50
DTX2	01/10/00	1520	370	<2	3,890	<2	<2	15	<2	<2	4,840	7	36
DTX2	04/14/00	1345	400	<1	3,700	<2	2	2	4	<1	5,000	<10	36
DTX2	07/07/00	1050	460	<1	4,130	<2	2	15	E2	<1	5,450	4	36
DTX2	10/17/00	1005	510	<2	3,640	<2	<2	26	3	<2	5,200	9	34
DTX3	01/10/00	1035	<10	<1	<1	<2	<1	4	8	<1	2,070	2	20
DTX3	04/14/00	1015	<10	<1	<1	<2	<1	6	14	<1	2,770	<3	28
DTX3	07/10/00	1030	<30	<1	<1	<2	<1	9	14	<1	3,070	<5	28
DTX3	10/17/00	1030	<30	<1	E.1	<2	1	2	17	<1	2,910	2	31
DTX4	01/06/00	0955	<10	<2	37	<2	<2	<1	E2	<2	4,180	3	32
DTX4	04/14/00	1145	<30	<2	33	<2	<2	20	3	<2	4,470	<11	35
DTX4	07/11/00	1010	<30	<1	36	<2	<2	10	E2	<2	4,200	3	27
DTX4	10/17/00	1215	<30	<1	69	<2	1	8	12	<2	4,300	5	32

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

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Well number	Date (mm/dd/yy)	Time (hh:mm)	Iron, dissolved (µg/L as Fe)	Lead, dissolved (µg/L as Pb)	Manganese, dissolved (µg/L as Mn)	Mercury, dissolved (µg/L as Hg)	Molybdenum, dissolved (µg/L as Mo)	Nickel, dissolved (µg/L as Ni)	Selenium, dissolved (µg/L as Se)	Silver, dissolved (µg/L as Ag)	Strontium, dissolved (µg/L as Sr)	Zinc, dissolved (µg/L as Zn)	Uranium, natural (µg/L as U)
DTX5	01/06/00	1310	100	<2	157	<0.2	<2	146	2	<2	5,270	3	39
DTX5	04/11/00	1345	120	<2	140	<2	<2	25	<2	<2	5,080	<14	38
DTX5	07/11/00	1150	90	<2	120	<2	<2	10	<2	<2	5,110	3	34
DTX5	10/16/00	1015	160	<1	126	<2	1	9	3	<2	4,540	5	35
DTX6	01/06/00	1530	<10	<2	<2	<2	<2	140	4	<2	5,240	5	37
DTX6	04/11/00	1000	<30	<1	<1	<2	<1	<1	6	<1	5,360	<11	38
DTX6	07/13/00	1300	<30	<1	<1	<2	<1	<1	E2	<1	5,720	5	41
DTX6	10/11/00	1400	<30	<1	1	<2	1	3	4	<2	5,370	7	38
DTX8A	01/13/00	1315	80	<1	180	<2	<1	6	<2	<1	2,460	2	<1
DTX8A	04/13/00	1415	120	<1	181	<2	<1	<1	<2	<1	2,320	<3	<1
DTX8A	07/10/00	1505	170	<1	181	<2	<1	3	<2	<1	2,520	<4	<1
DTX8A	10/16/00	1435	220	<1	172	<2	1	2	<2	<1	2,420	2	<1
DTX10A	01/11/00	1130	3,850	<1	158	<0.2	<1	9	<2	<1	5,450	2	<1
DTX10A	04/17/00	1415	3,970	<2	327	<2	<2	21	<2	<2	5,430	<12	<2
DTX10A	07/13/00	1020	4,130	<1	290	<2	1	3	<2	<1	5,490	4	<1
DTX10A	10/18/00	1040	4,110	<1	294	<2	1	1	<2	<2	5,510	4	M

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

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Well number	Date (mm/dd/yy)	Time (hh:mm)	Iron, dissolved (µg/L as Fe)	Lead, dissolved (µg/L as Pb)	Manganese, dissolved (µg/L as Mn)	Mercury, dissolved (µg/L as Hg)	Molybdenum, dissolved (µg/L as Mo)	Nickel, dissolved (µg/L as Ni)	Selenium, dissolved (µg/L as Se)	Silver, dissolved (µg/L as Ag)	Strontium, dissolved (µg/L as Sr)	Zinc, dissolved (µg/L as Zn)	Uranium, natural (µg/L as U)
D6	07/11/00	1345	<200	<7	3,970	E2	<7	20	9	<7	16,500	42	151
D6	01/11/00	1355	<10	<1	1,571	<2	<1	6	11	<1	15,600	15	121
D6	04/13/00	0945	<500	<1	3,680	<2	3	9	E17	<1	17,500	63	164
D6	10/12/00	1300	<150	<1	3,760	<2	4	8	15	<1	16,500	22	169
D13	01/12/00	1005	20	<1	84	<2	1	10	<2	<1	1,010	2	6
D13	04/17/00	1115	10	<1	82	<2	1	4	<2	<1	968	<3	8
D13	07/07/00	1230	10	<1	76	<2	1	4	<2	<1	997	<1	7
D13	10/16/00	1215	<10	<1	48	<2	1	2	E2	<1	1,160	2	7
D17	01/07/00	1500	<10	<1	249	<2	6	2	8	<1	288	<1	5
D17	04/10/00	1415	<10	<1	213	<2	6	<1	9	<1	305	<1	6
D17	07/06/00	1320	<10	<1	298	<2	6	1	9	<1	340	<1	7
D17	10/10/00	1245	<10	<1	347	<2	6	1	8	<1	295	<1	4
D25	01/12/00	1245	<10	<1	3,100	<0.2	10	38	3	<1	3,230	8	35
D25	04/10/00	1615	<30	<1	3,120	<2	9	14	2	<2	3,200	10	37
D25	07/06/00	1500	<30	<1	3,290	<2	9	33	2	<1	3,370	5	41
D25	10/10/00	1455	<30	<2	3,790	<2	14	14	6	<2	3,550	6	60

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

[pS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; <, less than; ND, no data available; E, value estimated by laboratory; --, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day.]

Well number	Date (mm/dd/yy)	Time (hh:mm)	Iron, dissolved (µg/L as Fe)	Lead, dissolved (µg/L as Pb)	Manganese, dissolved (µg/L as Mn)	Mercury, dissolved (µg/L as Hg)	Molybdenum, dissolved (µg/L as Mo)	Nickel, dissolved (µg/L as Ni)	Selenium, dissolved (µg/L as Se)	Silver, dissolved (µg/L as Ag)	Strontium, dissolved (µg/L as Sr)	Zinc, dissolved (µg/L as Zn)	Uranium, natural (µg/L as U)
D29	01/07/00	1130	7,210	<2	817	<2	<2	<1	<2	<2	5,830	8	<2
D29	04/10/00	1215	5,970	<1	813	<2	1	7	E2	<2	5,840	19	<1
D29	07/06/00	1105	7,080	<1	788	<2	<1	24	E1	<1	5,810	9	<1
D29	10/12/00	1110	6,640	<1	868	<2	1	7	3	<1	5,740	13	<1
D30	01/11/00	1600	50	<1	149	<2	2	15	<2	<1	5,880	4	17
D30	04/13/00	1200	E40	<1	293	<2	3	2	5	<1	6,010	<16	40
D30	07/10/00	1245	100	<1	253	<2	3	14	<2	<1	5,740	<11	35
D30	10/12/00	1450	1560	<1	260	<2	2	4	3	<1	5,780	6	35

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

[aSi/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; <, less than; ND, no data available; E, value estimated by laboratory; --, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Well number	Date (mm/dd/yy)	Time (hh:mm)	Plutonium-238, whole water (pCi/L)	Analytical uncertainty, Plutonium-238 (pCi/L)	MDC, Plutonium-238 (pCi/L)	Plutonium-239+240, whole water (pCi/L)	Analytical uncertainty, Plutonium-239+240, (pCi/L)	MDC, Plutonium-239+240 (pCi/L)
DTX1	01/10/00	1345	0	0.012	0.013	-0.002	0.004	0.027
DTX1	04/14/00	1220	--	--	--	--	--	--
DTX1	07/07/00	0940	--	--	--	--	--	--
DTX1	10/17/00	1200	0	0.012	0.014	0.005	0.010	0.014
DTX2	01/10/00	1520	0.008	0.016	0.022	0	0.020	0.022
DTX2	04/14/00	1345	--	--	--	--	--	--
DTX2	07/07/00	1050	--	--	--	--	--	--
DTX2	10/17/00	1005	0	0.007	0.008	0	0.007	0.008
DTX3	01/10/00	1035	0	0.004	0.004	0.001	0.004	0.009
DTX3	04/14/00	1015	--	--	--	--	--	--
DTX3	07/10/00	1030	--	--	--	--	--	--
DTX3	10/17/00	1030	0	0.008	0.009	0.002	0.007	0.018
DTX4	01/06/00	0955	-0.003	0.008	0.027	0.006	0.008	0.008
DTX4	04/14/00	1145	--	--	--	--	--	--
DTX4	07/14/00	1010	--	--	--	--	--	--
DTX4	10/17/00	1215	-0.002	0.003	0.024	0.009	0.012	0.012

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

[pCi/cm, picocuries per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; <, less than; ND, no data available; E, value estimated by laboratory; --, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Well number	Date (mm/dd/yy)	Time (hh:mm)	Plutonium-238, whole water (pCi/L)	Analytical uncertainty, Plutonium-238 (pCi/L)	MDC, Plutonium-238 (pCi/L)	Plutonium-239+240, whole water (pCi/L)	Analytical uncertainty, Plutonium-239+240, (pCi/L)	MDC, Plutonium-239+240 (pCi/L)
DTX5	01/06/00	1310	0	0.013	0.014	0.003	0.012	0.030
DTX5	04/11/00	1345	--	--	--	--	--	--
DTX5	07/11/00	1150	--	--	--	--	--	--
DTX5	10/16/00	1015	-0.001	0.002	0.016	0.003	0.006	0.008
DTX6	01/06/00	1530	-0.003	0.006	0.042	0	0.018	0.020
DTX6	04/14/00	1000	--	--	--	--	--	--
DTX6	07/13/00	1300	--	--	--	--	--	--
DTX6	10/11/00	1400	0	0.004	0.004	0.002	0.003	0.004
DTX8A	01/13/00	1315	0	0.006	0.006	-0.002	0.003	0.017
DTX8A	04/13/00	1415	--	--	--	--	--	--
DTX8A	07/10/00	1505	--	--	--	--	--	--
DTX8A	10/16/00	1435	0	0.004	0.005	-0.001	0.001	0.010
DTX10A	01/11/00	1130	0.003	0.012	0.030	0.003	0.012	0.030
DTX10A	04/17/00	1415	--	--	--	--	--	--
DTX10A	07/13/00	1020	--	--	--	--	--	--
DTX10A	10/18/00	1040	0.006	0.012	0.022	-0.001	0.010	0.030

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

[µS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; <, less than; ND, no data available; E, value estimated by laboratory; --, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Well number	Date (mm/dd/yy)	Time (hh:mm)	Plutonium-238, whole water (pCi/L)	Analytical uncertainty, Plutonium-238 (pCi/L)	MDC, Plutonium-238 (pCi/L)	Plutonium-238+240, whole water (pCi/L)	Analytical uncertainty, Plutonium-238+240 (pCi/L)	MDC, Plutonium-238+240 (pCi/L)
D6	01/11/00	1355	0.005	0.020	0.051	-0.004	0.007	0.051
D6	04/13/00	0945	--	--	--	--	--	--
D6	07/11/00	1345	--	--	--	--	--	--
D6	10/12/00	1300	0	0.007	0.008	0.001	0.007	0.020
D13	01/12/00	1005	-0.001	0.002	0.013	0.000	0.004	0.013
D13	04/17/00	1115	--	--	--	--	--	--
D13	07/07/00	1230	--	--	--	--	--	--
D13	10/16/00	1215	0.001	0.009	0.026	0.002	0.008	0.022
D17	01/07/00	1500	-0.001	0.001	0.008	0	0.004	0.004
D17	04/10/00	1415	--	--	--	--	--	--
D17	07/06/00	1320	--	--	--	--	--	--
D17	10/10/00	1245	-0.001	0.002	0.011	0.002	0.003	0.004
D25	01/12/00	1245	-0.004	0.006	0.034	0.001	0.011	0.034
D25	04/10/00	1615	--	--	--	--	--	--
D25	07/06/00	1500	--	--	--	--	--	--
D25	10/10/00	1455	0	0.008	0.009	0.003	0.006	0.009

Table 13. Water-quality data for monitoring wells near Deer Trail, Colorado, 2000—Continued

[pS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; <, less than; ND, no data available; E, value estimated by laboratory; --, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Well number	Date (mm/dd/yy)	Time (hh:mm)	Plutonium-238, whole water (pCi/L)	Analytical uncertainty, Plutonium-238 (pCi/L)	MDC, Plutonium-238 (pCi/L)	Plutonium-239+240, whole water (pCi/L)	Analytical uncertainty, Plutonium-239+240, (pCi/L)	MDC, Plutonium-239+240 (pCi/L)
D29	01/07/00	1130	-0.002	0.005	0.034	0.006	0.012	0.016
D29	04/10/00	1215	--	--	--	--	--	--
D29	07/06/00	1105	--	--	--	--	--	--
D29	10/12/00	1110	-0.010	0.010	0.051	-0.002	0.005	0.034
D30	01/11/00	1600	0	0.019	0.021	0.008	0.016	0.021
D30	04/13/00	1200	--	--	--	--	--	--
D30	07/10/00	1245	--	--	--	--	--	--
D30	10/12/00	1450	-0.002	0.004	0.031	-0.002	0.004	0.031

¹Value is significantly different from historical or subsequent data at the same site, and analytical bias is suspected. However, insufficient evidence from laboratory to reject or change value.

Table 14. Quality-control data for blank samples associated with ground-water samples collected near Deer Trail, Colorado, 2000

[$\mu\text{S}/\text{cm}$, microsiemens per centimeter at 25 degrees Celsius; mg/L , milligrams per liter; $\mu\text{g}/\text{L}$, micrograms per liter; pCi/L , picocuries per liter; Q, equipment blank; F, field blank; S, source-water blank; <, less than; E, value estimated by laboratory; --, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Blank type	Date (mm/dd/yy)	Time (hhmm)	Well number	Specific conductance, lab ($\mu\text{S}/\text{cm}$)	pH, laboratory (standard units)	Calcium, dissolved (mg/L as Ca)	Magnesium, dissolved (mg/L as Mg)	Potassium, dissolved (mg/L as K)	Sodium, dissolved (mg/L as Na)	Acid-neutralizing capacity, titration to pH 4.5 lab (mg/L as CaCO_3)	Chloride, dissolved (mg/L as Cl)
F	01/07/00	1430	D17	4	7.8	<0.02	<0.01	<0.2	<0.1	2	<0.3
F	01/11/00	1400	D6	E1	7.3	<0.02	<0.01	<0.2	<0.1	2	<0.3
F ¹	01/11/00	1405	D6	--	--	<0.01	<0.01	--	<0.1	--	--
F	04/11/00	1315	DTX5	4	7.4	<0.02	<0.01	<0.2	<0.1	2	<0.3
F	04/17/00	1045	D13	E2	8.0	<0.02	<0.01	<0.2	<0.1	2	<0.3
Q	04/21/00	1015	DTX10A	E2	7.5	.08	.02	<0.2	<0.1	2	<0.3
Q ¹	04/21/00	1016	DTX10A	--	--	.07	.01	--	<0.1	--	--
F	07/13/00	1240	DTX6	3	8.4	<0.02	<0.01	<0.2	<0.1	2	<0.3
S ¹	07/14/00	1045	DTX6	--	--	<0.01	<0.01	--	<0.1	--	--
S ¹	07/14/00	1050	None	--	--	<0.01	<0.01	--	<0.1	--	--
F	10/01/00	0940	DTX3	4	7.9	<0.01	<0.01	<0.2	<0.1	2	<0.1
F	10/17/00	1140	DTX1	4	8.1	<0.01	<0.01	<0.2	<0.1	1	<0.1

Table 14. Quality-control data for blank samples associated with ground-water samples collected near Deer Trail, Colorado, 2000—Continued

[µS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; Q, equipment blank; F, field blank; S, source-water blank; <, less than; E, value estimated by laboratory; —, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day.]

Blank type	Date (mm/dd/yyyy)	Fluoride, dissolved (mg/L as F)	Silica, dissolved (mg/L as SiO ₂)	Sulfate, dissolved (mg/L as SO ₄)	Nitrogen ammonia plus organic, dissolved (mg/L as N)	Nitrogen ammonia plus organic, total (mg/L as N)	Nitrogen ammonia, dissolved (mg/L as N)	Nitrite plus nitrate (mg/L as N)	Phosphorus, dissolved (mg/L as P)	Phosphorus, total (mg/L as P)
F	01/07/00	<0.1	<0.1	<0.3	E.10	<0.10	E.021	<0.04	<0.05	<0.05
F	01/11/00	<0.1	<0.1	<0.3	<0.10	<0.10	<0.029	E.0.02	<0.05	<0.05
F ¹	01/11/00	—	<0.1	—	—	—	—	—	—	—
F	04/11/00	<0.1	<0.1	<0.3	<0.10	<0.10	<0.029	.06	<0.05	<0.05
F	04/17/00	<0.1	<0.1	<0.3	<0.10	<0.12	<0.029	<0.04	<0.05	<0.05
Q	04/21/00	<0.1	E.04	<0.3	<0.10	<0.10	<0.029	<0.04	<0.05	<0.05
Q ¹	04/21/00	—	<0.1	—	—	—	—	—	—	—
F	07/13/00	<0.1	<0.1	<0.3	<0.10	<0.10	<0.029	<0.04	<0.05	<0.05
S ¹	07/14/00	—	<0.1	—	—	—	—	—	—	—
S ¹	07/14/00	—	<0.1	—	—	—	—	—	—	—
F	10/11/00	<0.2	<0.1	E.1	<0.10	<0.10	<0.049	<0.04	<0.05	<0.06
F	10/17/00	<0.2	<0.1	.2	<0.10	<0.10	<0.049	<0.04	<0.05	<0.06

Table 14. Quality-control data for blank samples associated with ground-water samples collected near Deer Trail, Colorado, 2000—Continued

[$\mu\text{S}/\text{cm}$, microsiemens per centimeter at 25 degrees Celsius; mg/L , milligrams per liter; $\mu\text{g}/\text{L}$, micrograms per liter; pCVL , picocuries per liter; Q , equipment blank; F , field blank; S , source-water blank; $<$, less than; E , value estimated by laboratory; $-$, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC , minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Blank type	Date (mm/dd/yy)	Chromium, dissolved ($\mu\text{g}/\text{L}$ as Cr)	Cobalt, dissolved ($\mu\text{g}/\text{L}$ as Co)	Copper, dissolved ($\mu\text{g}/\text{L}$ as Cu)	Iron, dissolved ($\mu\text{g}/\text{L}$ as Fe)	Lead, dissolved ($\mu\text{g}/\text{L}$ as Pb)	Manganese, dissolved ($\mu\text{g}/\text{L}$ as Mn)	Mercury, dissolved ($\mu\text{g}/\text{L}$ as Hg)	Molybdenum, dissolved ($\mu\text{g}/\text{L}$ as Mo)	Nickel, dissolved ($\mu\text{g}/\text{L}$ as Ni)	Selenium, dissolved ($\mu\text{g}/\text{L}$ as Se)
F	01/07/00	<0.8	<1	<1	<10	<1	<1	<0.2	<1	<1	<2
F	01/11/00	<8	<1	<1	<10	<1	<1	<2	<1	<1	<2
F ¹	01/11/00	<2	<1	<1	<3	<1	<1	-	<1	<1	-
F	04/11/00	<8	<1	<1	<10	<1	<1	<2	<1	<1	<2
F	04/17/00	<8	<1	<1	<10	<1	<1	<2	<1	<1	<2
Q	04/21/00	<8	<1	16	E10	<1	<1	<2	<1	<1	<2
Q ¹	04/21/00	<2	<1	16	10	<1	<1	-	<1	<1	-
F	07/13/00	E.6	<1	<1	<10	<1	<1	<2	<1	<1	<2
S ¹	07/14/00	<2	<1	<1	<3	<1	<1	-	<1	<1	-
S ¹	07/14/00	<2	<1	<1	<3	<1	<1	-	<1	<1	-
F	10/11/00	<8	<1	<1	<10	<1	<1	<2	<1	<1	<2
F	10/17/00	<8	<1	E.22	<10	<1	<1	<2	<1	<1	<2

Table 14. Quality-control data for blank samples associated with ground-water samples collected near Deer Trail, Colorado, 2000—Continued

[uS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; Q, equipment blank; F, field blank; S, source-water blank; <, less than; E, value estimated by laboratory; --, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Blank type	Date (mm/dd/yy)	Solids, residue on evaporation at 180°C, dissolved (mg/L)	Aluminum, dissolved (µg/L as Al)	Antimony, dissolved (µg/L as Sb)	Arsenic, dissolved (µg/L as As)	Barium, dissolved (µg/L as Ba)	Beryllium, dissolved (µg/L as Be)	Boron, dissolved (µg/L as B)	Bromide, dissolved (mg/L as Br)	Cadmium, dissolved (µg/L as Cd)
F	01/07/00	<10	2	<1	<2.0	<1	<1	<20	<0.01	<1.0
F	01/11/00	<10	<1	<1	<2.0	<1	<1	<20	<0.01	<1.0
F ¹	01/11/00	--	<1	<1	--	<1	<1	<2	--	<3
F	04/11/00	<10	<9	<1	<2.0	<1	<1	<20	<0.01	<1.0
F	04/17/00	<10	<1	<1	<2.0	<1	<1	<20	<0.01	<1.0
Q	04/21/00	<10	<1	<1	<2.0	<1	<1	<20	<0.01	<1.0
Q ¹	04/21/00	--	<1	<1	--	<1	<1	<2	--	<3
F	07/13/00	<10	<1	<1	<2.0	<1	<1	<20	<0.01	<1.0
S ¹	07/14/00	--	<1	<1	--	<1	<1	<2	--	<3
S ¹	07/14/00	--	<1	<1	--	<1	<1	<2	--	<3
F	10/11/00	<10	<1	<1	<2.0	<1	<1	<10	<0.01	<1
F	10/17/00	<10	<1	E.03	<2.0	<1	<1	<10	<0.01	<1

Table 14. Quality-control data for blank samples associated with ground-water samples collected near Deer Trail, Colorado, 2000—Continued

[µS/cm, microsiemens per centimeter at 25 degrees Celsius; mg/L, milligrams per liter; µg/L, micrograms per liter; pCi/L, picocuries per liter; Q, equipment blank; F, field blank; S, source-water blank; <, less than; E, value estimated by laboratory; --, no sample submitted; analytical uncertainty is the 2-sigma precision estimate calculated by the laboratory as a combined standard analytical uncertainty; MDC, minimum detectable concentration calculated for that sample; negative activity reported for some of the radiochemical samples means the sample counts were less than the laboratory background counts for that day]

Blank type	Date (mm/dd/yy)	Silver, dissolved (µg/L as Ag)	Strontium, dissolved (µg/L as Sr)	Thallium, dissolved (µg/L)	Zinc, dissolved (µg/L as Zn)	Uranium, natural (µg/L as U)	Plutonium-238 (pCi/L)	Analytical uncertainty, Plutonium-238 (pCi/L)	MDC, Plutonium-238 (pCi/L)	Plutonium-239+240 (pCi/L)	Analytical uncertainty, Plutonium-239+240 (pCi/L)	MDC, Plutonium-239+240 (pCi/L)
F	01/07/00	<1	<1.0	--	<1	<1	0	0.004	0.004	0.001	0.003	0.004
F	01/11/00	<1	<1.0	--	<1	<1	-0.001	0.001	0.008	0	0.004	0.004
F ¹	01/11/00	<1	<1	<1	<1	<1	--	--	--	--	--	--
F	04/11/00	<1	<1.0	--	<1	<1	--	--	--	--	--	--
F	04/17/00	<1	<1.0	--	2	<1	--	--	--	--	--	--
Q	04/21/00	<1	E.8	--	<3	<1	20.012	0.008	0.009	0.001	0.002	0.004
Q ¹	04/21/00	<1	8	<1	1	<1	--	--	--	--	--	--
F	07/13/00	<1	<1.0	--	<1	<1	--	--	--	--	--	--
S ¹	07/14/00	<1	<1	<1	<1	<1	--	--	--	--	--	--
S ¹	07/14/00	<1	<1	<1	<1	<1	--	--	--	--	--	--
F	10/11/00	<1	<8	--	<1	<1	0.003	0.004	0.004	0.002	0.005	0.009
F	10/17/00	<1	<8	--	<1	<1	0.001	0.003	0.009	0	0.004	0.004

¹Blank sample for low-level trace-element analysis.

²Analytical bias is suspected. However, insufficient evidence from laboratory to reject or change value.

Table 15. Comparison of water-quality data for replicate and regular ground-water samples collected near Deer Trail, Colorado, 2000

[RPD, relative percent difference, which is defined as $[(\text{sample value} - \text{replicate value})/(\text{sample value} + \text{replicate value})/2] \times 100$; relative percent difference was calculated on unrounded data; --, not analyzed; ND, not determined because data were less than the minimum reporting limit or estimated; <, less than; E, value estimated by laboratory]

Property or constituent	DTX4				D29				DTX3			
	01/06/00				01/07/00				01/10/00			
	Date	Time	Sample	Replicate	RPD	Sample	Replicate	RPD	Sample	Replicate	RPD	RPD
Specific conductance, lab ($\mu\text{S}/\text{cm}$ at 25°C)			2,950	2,960	0	4,020	4,010	0	1,410	1,410	0	0
pH, WHI laboratory (standard units)			7.0	7.0	0	6.9	6.8	1	7.4	7.4	0	0
Hardness total (mg/L as CaCO_3)			1,700	1,700	0	2,800	2,700	4	720	660	9	9
Calcium, dissolved (mg/L as Ca)			531	523	2	535	526	2	170	158	7	7
Magnesium, dissolved (mg/L as Mg)			86.9	84.0	3	346	327	6	69.9	63.6	9	9
Sodium, dissolved (mg/L as Na)			160	158	1	143	140	2	58.2	54.0	8	8
Potassium, dissolved (mg/L as K)			8.8	8.9	-1	11.1	11.6	-4	6.2	6.0	3	3
Acid neutralizing capacity, titration to 4.5, lab (mg/L as CaCO_3)			353	393	-11	293	292	0	274	274	0	0
Sulfate, dissolved (mg/L as SO_4)			1,590	1,620	-2	2,690	2,710	-1	516	516.86	0	0
Chloride, dissolved (mg/L as Cl)			15.2	15.6	-3	11.5	11.6	-1	17.1	17.2	-1	-1
Fluoride, dissolved (mg/L as F)			2	2	6	5	4	13	4	4	0	0
Bromide, dissolved (mg/L as Br)			22	22	0	18	2	-5	22	22	0	0
Silica, dissolved (mg/L as SiO_2)			11.6	11.8	-1	21.1	20.8	1	15.7	15.4	2	2
Solids, residue on evaporation at 180°C , dissolved (mg/L)			2,800	2,850	-2	4,260	4,270	0	1,090	1,090	0	0
Dissolved solids, sum of constituents (mg/L)			2,620	2,670	-2	3,950	3,930	1	1,030	1,010	2	2
Nitric plus nitrate (mg/L as N)			33	32	3	E.02	E.02	0	2.78	2.78	0	0
Nitrogen ammonia, dissolved (mg/L as N)			.063	.09	-31	.446	.446	0	E.014	<.029	ND	ND
Nitrogen ammonia plus organic, total (mg/L as N)			.29	.30	-2	.51	.50	2	.16	.14	10	10
Nitrogen ammonia plus organic, dissolved (mg/L as N)			.26	.25	6	.45	.48	-6	.19	.14	32	32
Phosphorus, total (mg/L as P)			<.05	<.05	0	E.04	<.05	ND	<.05	<.05	0	0
Phosphorus, dissolved (mg/L as P)			<.05	<.05	0	<.05	E3.04	ND	<.05	<.05	0	0

Table 15. Comparison of water-quality data for replicate and regular ground-water samples collected near Deer Trail, Colorado, 2000—Continued

[RPD, relative percent difference, which is defined as $[(\text{sample value} - \text{replicate value}) / ((\text{sample value} + \text{replicate value}) / 2)] \times 100$; relative percent difference was calculated on unrounded data; -, not analyzed; ND, not determined because data were less than the minimum reporting limit or estimated; <, less than; E, value estimated by laboratory]

Property or constituent	Well number			DTX4			D29			DTX3		
	Date			01/06/00			01/07/00			01/10/00		
	Time	Sample	Replicate	Sample	Replicate	RPD	Sample	Replicate	RPD	Sample	Replicate	RPD
Aluminum, dissolved ($\mu\text{g/L}$ as Al)		<2	<2			0	6	5	18	2	<1	ND
Antimony, dissolved ($\mu\text{g/L}$ as Sb)		<2	<2			0	<2	<2	0	<1	<1	0
Arsenic, dissolved ($\mu\text{g/L}$ as As)		E1	<2			ND	<2	<2	0	<2	<2	0
Barium, dissolved ($\mu\text{g/L}$ as Ba)		14	14			0	9	9	0	23	23	0
Beryllium, dissolved ($\mu\text{g/L}$ as Be)		<2	<2			0	<2	<2	0	<1	<1	0
Boron, dissolved ($\mu\text{g/L}$ as B)		290	280			3	180	190	-3	210	210	0
Cadmium, dissolved ($\mu\text{g/L}$ as Cd)		<2.0	<2.0			0	<2.0	<2.0	0	<1	<1	0
Chromium, dissolved ($\mu\text{g/L}$ as Cr)		<8	E.7			ND	<1.0	<1.0	0	E.6	<8	ND
Cobalt, dissolved ($\mu\text{g/L}$ as Co)		<1	<1			0	<2	<2	0	<1	<1	0
Copper, dissolved ($\mu\text{g/L}$ as Cu)		5	5			0	6	7	-15	3	2	40
Iron, dissolved ($\mu\text{g/L}$ as Fe)		<10	E9			ND	7,210	7,110	1	<10	<10	0
Lead, dissolved ($\mu\text{g/L}$ as Pb)		<2	<2			0	<2	<2	0	<1	<1	0
Manganese, dissolved ($\mu\text{g/L}$ as Mn)		37	36			3	817	818	0	<1	<1	0
Mercury, dissolved ($\mu\text{g/L}$ as Hg)		<2	<2			0	<2	<2	0	<2	<2	0
Molybdenum, dissolved ($\mu\text{g/L}$ as Mo)		<2	<2			0	<2	<2	0	<1	<1	0
Nickel, dissolved ($\mu\text{g/L}$ as Ni)		<1	<1			0	<1	<1	0	4	4	0
Selenium, dissolved ($\mu\text{g/L}$ as Se)		E2	E2			0	<2	<2	0	8	8	0
Silver, dissolved ($\mu\text{g/L}$ as Ag)		<2	<2			0	<2	<2	0	<1	<1	0
Strontium, dissolved ($\mu\text{g/L}$ as Sr)		4,180	4,120			1	5,830	5,760	1	2,070	2,010	3
Zinc, dissolved ($\mu\text{g/L}$ as Zn)		3	3			0	8	8	0	2	2	0
Uranium, natural ($\mu\text{g/L}$ as U)		32	32			0	<2	<2	0	20	20	0
Plutonium-238, whole water (pCi/L as Pu)		-0.003	0			200	0.002	0	200	0	0.001	-200
Plutonium-239-240, whole water (pCi/L as Pu)		0.006	0			200	0.006	0	200	0.001	-0.001	ND

Table 15. Comparison of water-quality data for replicate and regular ground-water samples collected near Deer Trail, Colorado, 2000—Continued

[RPD, relative percent difference, which is defined as $[(\text{sample value} - \text{replicate value})/(\text{sample value} + \text{replicate value})/2] \times 100$; relative percent difference was calculated on unrounded data; --, not analyzed; ND, not determined because data were less than the minimum reporting limit or estimated; <, less than; E, value estimated by laboratory]

Property or constituent	Well number								
	DTX3			D6					
	Date	04/11/00		04/13/00		D25			
Time	1015	1016		0945	0950	1500	1505		
Sample	Sample	Replicate	RPD	Sample	Replicate	RPD	Sample		
Replicate	Replicate	Replicate	Replicate	Replicate	Replicate	Replicate	Replicate		
Specific conductance, lab (µS/cm at 25 °C)	1,880	1,880	0	16,300	16,300	0	4,510	4,490	0
pH, WHI, laboratory (standard units)	7.3	7.3	0	7.2	7.2	0	7.3	7.3	0
Hardness total (mg/L as CaCO ₃)	990	960	3	11,000	10,000	10	2,600	2,500	4
Calcium, dissolved (mg/L as Ca)	240	232	3	446	431	3	638	624	2
Magnesium, dissolved (mg/L as Mg)	94.2	91.1	3	2,320	2,230	4	232	223	4
Sodium, dissolved (mg/L as Na)	72.8	72.3	1	2,100	2,000	5	286	271	5
Potassium, dissolved (mg/L as K)	7.3	7.0	4	12.9	12.6	3	7.3	6.0	20
Acid neutralizing capacity, titration to 4.5, lab (mg/L as CaCO ₃)	314	314	0	640	639	0	496	486	2
Sulfate, dissolved (mg/L as SO ₄)	770	762	1	12,900	13,100	-2	2,520	2,530	0
Chloride, dissolved (mg/L as Cl)	26.0	25.1	4	380	384	-1	99.8	100	0
Fluoride, dissolved (mg/L as F)	.4	.4	0	.8	.7	7	1.0	1.0	3
Bromide, dissolved (mg/L as Br)	.28	.28	0	4.32	4.44	-3	10.05	10.06	-18
Silica, dissolved (mg/L as SiO ₂)	16.6	16.0	4	21.3	21.2	0	28.5	27.7	3
Solids, residue on evaporation at 180°C, dissolved (mg/L)	1,550	1,530	1	20,300	20,800	-2	4,640	4,690	-1
Dissolved solids, sum of constituents (mg/L)	1,440	1,410	2	18,600	18,700	-1	4,130	4,090	1
Nitrite plus nitrate (mg/L as N)	3.80	3.80	0	13.1	12.4	5	3.0	3.0	0
Nitrogen ammonia, dissolved (mg/L as N)	<.029	<.029	0	.045	.059	-27	.138	.148	-7
Nitrogen ammonia plus organic, total (mg/L as N)	.17	.16	7	1.4	1.2	15	.77	.79	-3
Nitrogen ammonia plus organic, dissolved (mg/L as N)	<.10	<.10	0	1.5	1.5	0	.74	.70	6
Phosphorus, total (mg/L as P)	<.05	<.05	0	E.04	E.03	ND	.15	.15	0
Phosphorus, dissolved (mg/L as P)	<.05	<.05	0	E.04	<.05	ND	.15	.15	0

Table 15. Comparison of water-quality data for replicate and regular ground-water samples collected near Deer Trail, Colorado, 2000—Continued

[RPD, relative percent difference, which is defined as $[(\text{sample value} - \text{replicate value})/(\text{sample value} + \text{replicate value})/2] \times 100$; relative percent difference was calculated on unrounded data; --, not analyzed; ND, not determined because data were less than the minimum reporting limit or estimated; <, less than; E, value estimated by laboratory]

Property or constituent	Well number		DTX3		D6		D25	
	Date		04/11/00		04/13/00		07/06/00	
	Time		Sample	Replicate	RPD	Sample	Replicate	RPD
Aluminum, dissolved ($\mu\text{g/L}$ as Al)		<6		<1	ND	1	1	0
Antimony, dissolved ($\mu\text{g/L}$ as Sb)		<1		<1	0	<1	<1	0
Arsenic, dissolved ($\mu\text{g/L}$ as As)		<2		<2	0	E9	E7	ND
Barium, dissolved ($\mu\text{g/L}$ as Ba)		23		23	0	6	6	0
Beryllium, dissolved ($\mu\text{g/L}$ as Be)		<1		<1	0	<1	<1	0
Boron, dissolved ($\mu\text{g/L}$ as B)		220		220	0	E750	750	ND
Cadmium, dissolved ($\mu\text{g/L}$ as Cd)		<1.0		<1.0	0	<1.0	<1.0	0
Chromium, dissolved ($\mu\text{g/L}$ as Cr)		<8		<1	ND	<8	<1.0	ND
Cobalt, dissolved ($\mu\text{g/L}$ as Co)		<1		<1	0	6	6	0
Copper, dissolved ($\mu\text{g/L}$ as Cu)		3		3	0	61	63	-3
Iron, dissolved ($\mu\text{g/L}$ as Fe)		<10		<10	0	<500	<500	0
Lead, dissolved ($\mu\text{g/L}$ as Pb)		<1		<1	0	<1	<1	0
Manganese, dissolved ($\mu\text{g/L}$ as Mn)		<1		<1	0	3,680	3,770	-2
Mercury, dissolved ($\mu\text{g/L}$ as Hg)		<2		<2	0	<2	<2	0
Molybdenum, dissolved ($\mu\text{g/L}$ as Mo)		<1		<1	0	3	3	0
Nickel, dissolved ($\mu\text{g/L}$ as Ni)		6		6	0	9	9	0
Selenium, dissolved ($\mu\text{g/L}$ as Se)		14		13	5	E17	E5	ND
Silver, dissolved ($\mu\text{g/L}$ as Ag)		<1		<1	0	<1	<1	0
Strontium, dissolved ($\mu\text{g/L}$ as Sr)		2,770		2,800	-1	17,500	16,900	3
Zinc, dissolved ($\mu\text{g/L}$ as Zn)		<3		<1	ND	63	66	-5
Uranium, natural ($\mu\text{g/L}$ as U)		28		28	0	164	164	0
Plutonium-238, whole water (pCi/L as Pu)		--		--	--	--	--	--
Plutonium-239-240, whole water (pCi/L as Pu)		--		--	--	--	--	--

Table 15. Comparison of water-quality data for replicate and regular ground-water samples collected near Deer Trail, Colorado, 2000—Continued

[RPD, relative percent difference, which is defined as [(sample value-replicate value)/(sample value + replicate value)(2)] × 100; relative percent difference was calculated on unrounded data; --, not analyzed; ND, not determined because data were less than the minimum reporting limit or estimated; <, less than; E, value estimated by laboratory]

Property or constituent	D6				D25				D6			
	Well number		Date		10/10/00		10/12/00		10/12/00		10/12/00	
	1345	1350	Replicate	RPD	Sample	Replicate	Sample	Replicate	RPD	Sample	Replicate	RPD
Specific conductance, lab (µS/cm at 25 °C)	16,000	16,000		0	4,900	4,910			0	16,200	16,100	1
pH, WHI laboratory (standard units)	7.2	7.2		0	7.2	7.2			0	7.1	7.1	0
Hardness total (mg/L as CaCO ₃)	10,000	10,000		0	2,900	2,900			0	10,000	10,000	0
Calcium, dissolved (mg/L as Ca)	414	417		-1	794	797			0	437	439	0
Magnesium, dissolved (mg/L as Mg)	2,170	2,180		0	219	220			0	2,190	2,200	0
Sodium, dissolved (mg/L as Na)	2,070	2,070		0	334	334			0	2,110	2,100	0
Potassium, dissolved (mg/L as K)	12.0	12.7		-5	8.3	8.2			1	10.4	10.0	4
Acid neutralizing capacity, titration to 4.5, lab (mg/L as CaCO ₃)	638	639		0	585	473			21	640	640	0
Sulfate, dissolved (mg/L as SO ₄)	13,100	13,200		-1	2,860	2,850			0	13,100	13,300	-2
Chloride, dissolved (mg/L as Cl)	404	403		0	81.5	79.1			3	419	418	0
Fluoride, dissolved (mg/L as F)	.8	.7		9	1.0	1.0			0	.8	.8	0
Bromide, dissolved (mg/L as Br)	3.95	3.94		0	1.38	1.39			-1	4.15	4.14	0
Silica, dissolved (mg/L as SiO ₂)	20.6	20.9		-1	28.8	29.0			-1	19.8	19.6	1
Solids, residue on evaporation at 180°C, dissolved (mg/L)	21,100	21,200		0	5,270	5,170			2	21,500	20,900	3
Dissolved solids, sum of constituents (mg/L)	18,700	18,700		0	4,710	4,620			2	18,700	19,000	-2
Nitrite plus nitrate (mg/L as N)	14.4	15.5		-7	4.9	5.0			-2	14.6	14.6	0
Nitrogen ammonia, dissolved (mg/L as N)	<.030	<.030		0	.10	.11			0	.131	.135	-3
Nitrogen ammonia plus organic, total (mg/L as N)	1.4	1.4		0	.88	.73			19	1.5	1.5	0
Nitrogen ammonia plus organic, dissolved (mg/L as N)	1.5	1.5		0	1.11	.83			-153	1.1	.98	12
Phosphorus, total (mg/L as P)	E.04	<.05		ND	.12	.11			5	E.04	<.05	ND
Phosphorus, dissolved (mg/L as P)	E.03	E.04		ND	.11	.11			0	<.05	E.04	ND
Aluminum, dissolved (µg/L as Al)	<1	<1		0	<2	<2			0	<1	1	ND

Table 16. Statistical comparison of median concentrations for selected chemical constituents in ground-water samples collected near Deer Trail, Colorado, 1999-2000, and lowest applicable water-quality standard

[--, not computed or not applicable ; standard is from Colorado Department of Public Health and Environment (1997); H, health-based standard; A, agricultural standard; E, value estimated by laboratory; *, value estimated because variance was zero]

Well	Sample size for statistical comparison	Minimum ground-water value used for statistical comparison ¹	Maximum ground-water value used for statistical comparison ¹	Median ground-water value used for statistical comparison ¹	Colorado standard	Type of regulatory standard	Probability that the median concentration exceeded the regulatory standard ²
Nitrate ³ (mg/L as N)							
D6	8	11	14.6	12	10	H	0.9961
D13	7	.018	.049	.025	10	H	.0078
D17	8	.679	5.68	2.60	10	H	.0039
D25	7	2.2	7.4	3.4	10	H	.0078
D29	8	.018	.03	.023	10	H	.0039
D30	8	.018	.06	.018	10	H	.0039
DTX1	7	1.1	2.8	1.29	10	H	.0078
DTX10A	8	.018	.025	.018	10	H	.0039
DTX2	8	.018	.05	.022	10	H	.0039
DTX3	8	.018	4.3	3.3	10	H	.0039
DTX4	8	.018	6.52	.241	10	H	.0039
DTX5	8	.018	.1	.018	10	H	.0039
DTX6	8	.22	.32	.25	10	H	.0039
DTX8A	8	.018	.06	.018	10	H	.0039
Arsenic ⁴ (µg/L as As)							
D6	8	1	3.2	2.2	5	H	.0039
D13	8	.05	1	1	5	H	.0039
D17	8	1	2	1.35	5	H	.0039
D25	8	2	6	2.7	5	H	.0352
D29	8	.05	2	1	5	H	.0039
D30	8	.05	2.7	1	5	H	.0039
DTX1	8	1	3.5	1.95	5	H	.0039
DTX10A	8	.05	1	1	5	H	.0039
DTX2	8	.05	2	1.15	5	H	.0039
DTX3	8	.05	1	1	5	H	.0039
DTX4	8	.05	2	1	5	H	.0039
DTX5	8	.05	1	1	5	H	.0039
DTX6	8	.05	2.2	1	5	H	.0039
DTX8A	8	.05	1	1	5	H	.0039
Cadmium ⁵ (µg/L as Cd)							
D6	8	.1	3.5	2	5	H	.0039
D13	8	0.022	0.5	0.5	5	H	.0039
D17	8	.05	.5	.5	5	H	.0039

Table 16. Statistical comparison of median concentrations for selected chemical constituents in ground-water samples collected near Deer Trail, Colorado, 1999-2000, and lowest applicable water-quality standard—Continued

[--, not computed or not applicable; standard is from Colorado Department of Public Health and Environment (1997); H, health-based standard; A, agricultural standard; E, value estimated by laboratory; *, value estimated because variance was zero]

Well	Sample size for statistical comparison	Minimum ground-water value used for statistical comparison ¹	Maximum ground-water value used for statistical comparison ¹	Median ground-water value used for statistical comparison ¹	Colorado standard	Type of regulatory standard	Probability that the median concentration exceeded the regulatory standard ²
Cadmium ⁵ (µg/L as Cd)--Continued							
D25	8	.5	1.5	1	5	H	.0039
D29	8	.038	1	1	5	H	.0039
D30	8	.037	1	.5	5	H	.0039
DTX1	8	.5	1	1	5	H	.0039
DTX10A	8	.05	1	.75	5	H	.0039
DTX2	8	.5	1	1	5	H	.0039
DTX3	8	.05	.5	.5	5	H	.0039
DTX4	8	.1	1	1	5	H	.0039
DTX5	8	.05	1	1	5	H	.0039
DTX6	8	.1	1	1	5	H	.0039
DTX8A	8	.05	.5	.5	5	H	.0039
Chromium (µg/L as Cr)							
D6	8	.4	18	.5	100	H, A	.0039
D13	8	.4	11	.45	100	H, A	.0039
D17	8	.4	3	.4	100	H, A	.0039
D25	8	.4	27	1.55	100	H, A	.0039
D29	8	.4	18	.5	100	H, A	.0039
D30	8	.4	9.5	.95	100	H, A	.0039
DTX1	8	.4	14	.5	100	H, A	.0039
DTX10A	8	.4	5.7	.75	100	H, A	.0039
DTX2	8	.4	11	.7	100	H, A	.0039
DTX3	8	.4	13	.5	100	H, A	.0039
DTX4	8	.4	11	.75	100	H, A	.0039
DTX5	8	.4	8.2	.95	100	H, A	.0039
DTX6	8	.4	11	1.1	100	H, A	.0039
DTX8A	8	.4	.8	.45	100	H, A	.0039

Table 16. Statistical comparison of median concentrations for selected chemical constituents in ground-water samples collected near Deer Trail, Colorado, 1999-2000, and lowest applicable water-quality standard—Continued

[--, not computed or not applicable ; standard is from Colorado Department of Public Health and Environment (1997); H, health-based standard; A, agricultural standard; E, value estimated by laboratory; *, value estimated because variance was zero]

Well	Sample size for statistical comparison	Minimum ground-water value used for statistical comparison ¹	Maximum ground-water value used for statistical comparison ¹	Median ground-water value used for statistical comparison ¹	Colorado standard	Type of regulatory standard	Probability that the median concentration exceeded the regulatory standard ²
Copper (µg/L as Cu)							
D6	8	5	61	31	200	A	.0039
D13	8	2	4	2.5	200	A	.0039
D17	8	.05	1	.275	200	A	.0039
D25	8	7	11	9	200	A	.0039
D29	8	6	10	7.5	200	A	.0039
D30	8	5	18	8.5	200	A	.0039
DTX1	8	6	14	8.5	200	A	.0039
DTX10A	8	3	6	5.5	200	A	.0039
DTX2	8	5	11	7	200	A	.0039
DTX3	8	2	5	3	200	A	.0039
DTX4	8	4	8	6	200	A	.0039
DTX5	8	4	8	6.5	200	A	.0039
DTX6	8	6	16	8.5	200	A	.0039
DTX8A	8	2	4	2	200	A	.0039
Lead ⁶ (µg/L as Pb)							
D6	8	.5	3.5	2	50	H	.0039
D13	8	.5	.5	.5	50	H	.0039*
D17	8	.5	.5	.5	50	H	.0039*
D25	8	.5	1.5	1	50	H	.0039
D29	8	.5	1	1	50	H	.0039
D30	8	.5	1	.5	50	H	.0039
DTX1	8	.5	1	1	50	H	.0039
DTX10A	8	.5	1	.75	50	H	.0039
DTX2	8	.5	1	1	50	H	.0039
DTX3	8	.5	.5	.5	50	H	.0039*
DTX4	8	.5	1	1	50	H	.0039
DTX5	8	.5	1	1	50	H	.0039
DTX6	8	.5	1	1	50	H	.0039
DTX8A	8	.5	.5	.5	50	H	.0039*

Table 16. Statistical comparison of median concentrations for selected chemical constituents in ground-water samples collected near Deer Trail, Colorado, 1999-2000, and lowest applicable water-quality standard—Continued

[--, not computed or not applicable; standard is from Colorado Department of Public Health and Environment (1997); H, health-based standard; A, agricultural standard; E, value estimated by laboratory; *, value estimated because variance was zero]

Well	Sample size for statistical comparison	Minimum ground-water value used for statistical comparison ¹	Maximum ground-water value used for statistical comparison ¹	Median ground-water value used for statistical comparison ¹	Colorado standard	Type of regulatory standard	Probability that the median concentration exceeded the regulatory standard ²
Mercury (µg/L as Hg)							
D6	8	.05	.2	.1	2	H	.0039
D13	8	.05	.1	.1	2	H	.0039
D17	8	.05	.1	.1	2	H	.0039
D25	8	.05	.1	.1	2	H	.0039
D29	8	.05	.1	.1	2	H	.0039
D30	8	.05	.1	.1	2	H	.0039
DTX1	8	.05	.1	.1	2	H	.0039
DTX10A	8	.05	.1	.1	2	H	.0039
DTX2	8	.05	.1	.1	2	H	.0039
DTX3	8	0.05	0.1	0.1	2	H	.0039
DTX4	8	.05	.1	.1	2	H	.0039
DTX5	8	.05	.1	.1	2	H	.0039
DTX6	8	.05	.1	.1	2	H	.0039
DTX8A	8	.05	.1	.1	2	H	.0039
Molybdenum ⁷ (µg/L as Mo)							
D6	8	.5	4	3.25	.. ⁶	--	--
D13	8	.5	1	1	.. ⁶	--	--
D17	8	6	7	6	.. ⁶	--	--
D25	8	9	14	10.5	.. ⁶	--	--
D29	8	.5	1	1	.. ⁶	--	--
D30	8	1	4	2.5	.. ⁶	--	--
DTX1	8	5	7	5.5	.. ⁶	--	--
DTX10A	8	.5	3	1	.. ⁶	--	--
DTX2	8	1	2	1	.. ⁶	--	--
DTX3	8	.5	1	1	.. ⁶	--	--
DTX4	8	1	1	1	.. ⁶	--	--
DTX5	8	1	1	1	.. ⁶	--	--
DTX6	8	.5	1	1	.. ⁶	--	--
DTX8A	8	.5	2	.5	.. ⁶	--	--

Table 16. Statistical comparison of median concentrations for selected chemical constituents in ground-water samples collected near Deer Trail, Colorado, 1999-2000, and lowest applicable water-quality standard—Continued

[--, not computed or not applicable ; standard is from Colorado Department of Public Health and Environment (1997); H, health-based standard; A, agricultural standard; E, value estimated by laboratory; *, value estimated because variance was zero]

Well	Sample size for statistical comparison	Minimum ground-water value used for statistical comparison ¹	Maximum ground-water value used for statistical comparison ¹	Median ground-water value used for statistical comparison ¹	Colorado standard	Type of regulatory standard	Probability that the median concentration exceeded the regulatory standard ²
Nickel (µg/L as Ni)							
D6	8	6	23	12	100	H	.0039
D13	8	2	10	4	100	H	.0039
D17	8	.5	2	1	100	H	.0039
D25	8	4	38	16.5	100	H	.0039
D29	8	.5	24	15	100	H	.0039
D30	8	2	15	12	100	H	.0039
DTX1	8	4	31	16	100	H	.0039
DTX10A	8	1	21	6.5	100	H	.0039
DTX2	8	2	26	14	100	H	.0039
DTX3	8	2	9	3.5	100	H	.0039
DTX4	8	.5	20	12	100	H	.0039
DTX5	8	6	46	11.5	100	H	.0039
DTX6	8	.5	40	6	100	H	.0039
DTX8A	8	0.5	6	3	100	H	.0039
Selenium (µg/L as Se)							
D6	8	6	17	8.4	20	A	.0039
D13	8	.5	1.7	1.1	20	A	.0039
D17	8	7.5	9	8.3	20	A	.0039
D25	8	1	6	2.7	20	A	.0039
D29	8	.5	3.3	1.35	20	A	.0039
D30	8	.5	4.9	1.2	20	A	.0039
DTX1	8	1	5.9	2.55	20	A	.0039
DTX10A	8	.5	2	1.2	20	A	.0039
DTX2	8	.5	4.3	1.55	20	A	.0039
DTX3	8	4	16.5	14.05	20	A	.0039
DTX4	8	1	11.6	2	20	A	.0039
DTX5	8	.5	3.3	1.2	20	A	.0039
DTX6	8	.5	6.1	3.35	20	A	.0039
DTX8A	8	.5	1.2	1.1	20	A	.0039

Table 16. Statistical comparison of median concentrations for selected chemical constituents in ground-water samples collected near Deer Trail, Colorado, 1999-2000, and lowest applicable water-quality standard—Continued

[--, not computed or not applicable; standard is from Colorado Department of Public Health and Environment (1997); H, health-based standard; A, agricultural standard; E, value estimated by laboratory; *, value estimated because variance was zero]

Well	Sample size for statistical comparison	Minimum ground-water value used for statistical comparison ¹	Maximum ground-water value used for statistical comparison ¹	Median ground-water value used for statistical comparison ¹	Colorado standard	Type of regulatory standard	Probability that the median concentration exceeded the regulatory standard ²
Zinc (µg/L as Zn)							
D6	8	5	63	30	2,000	A	.0039
D13	8	.5	3	1.75	2,000	A	.0039
D17	8	.5	.5	.5	2,000	A	.0039*
D25	8	5	25	7	2,000	A	.0039
D29	8	8	19	11	2,000	A	.0039
D30	8	4	10	6.5	2,000	A	.0039
DTX1	8	4	8	6	2,000	A	.0039
DTX10A	8	2	10	4.5	2,000	A	.0039
DTX2	8	4	9	6	2,000	A	.0039
DTX3	8	1	2.5	2	2,000	A	.0039
DTX4	8	3	6	4	2,000	A	.0039
DTX5	8	3	7	4	2,000	A	.0039
DTX6	8	5	7	6	2,000	A	.0039
DTX8A	8	1.5	2	2	2,000	A	.0039

¹Many of the values used in the statistical comparison were derived from concentrations that are less than the minimum reporting limit and therefore were set equal to one-half the minimum reporting limit.

²Value is 1 minus the p-value resulting from a one-tailed Sign Test (Helsel and Hirsch, 1995), which is used to indicate the level of statistical evidence that selected constituent concentrations are significantly greater than regulatory standards. A value close to 1.0 indicates more evidence that the median concentration exceeded the standard, whereas a value close to 0 indicates little evidence that the median concentration exceeded the standard. The percent confidence of the test can be determined by subtracting the p-value from 1 and multiplying by 100. For example, if the p-value is 0.100, 1-p is 0.900, so the median concentration is greater than the regulatory standard with 90-percent confidence.

³Data compared to standard are for nitrite plus nitrate. Results indicate nitrite is a minor component.

⁴During 2000, standard was a proposed maximum contaminant level.

⁵All data were less than laboratory minimum reporting limit, except for one value (D6). Minimum reporting limits were sometimes greater than the water-quality standard.

⁶All data were less than laboratory minimum reporting limit. Laboratory minimum reporting limit is less than the water-quality standard.

⁷No regulatory standard for this constituent.

Table 17. Statistical evaluation of monotonic time-series trend using the Kendall's tau correlation coefficient for selected constituents in ground-water samples collected near Deer Trail, Colorado, 1999-2000

[Only values in bold indicate statistically significant trends. For this report, the Kendall's tau statistic (Helsel and Hirsch, 1995) is used as an indicator of monotonic correlation between concentration and time. By this method, positive values of Kendall's tau indicate upward trends and negative values indicate downward trends. Kendall's tau is a number between -1 and 1 that indicates increasing strength of the correlation. A p-value is listed to indicate the level of significance of the correlation; --, not computed]

Well	Nitrate		Arsenic		Cadmium		Chromium		Copper		Lead	
	Tau	p-value	Tau	p-value	Tau	p-value	Tau	p-value	Tau	p-value	Tau	p-value
D6	.786	.009	.107	.804	-.428	.174	-.643	.035	.214	.536	-- ¹	-- ¹
D13	.238	.548	.428	.174	-- ¹	-- ¹	-.571	.063	-.036	1.000	-- ¹	-- ¹
D17	.214	.536	-.428	.174	-- ¹	-- ¹	-.607	.048	.714	.019	-- ¹	-- ¹
D25	-.428	.230	-.071	.902	-- ¹	-- ¹	-.250	.458	.250	.458	-- ¹	-- ¹
D29	-- ¹	-- ¹	.071	.902	-- ¹	-- ¹	-.286	.386	-.036	1.000	-- ¹	-- ¹
D30	-.607	.048	.500	.108	-- ¹	-- ¹	-.500	.108	.071	.902	-- ¹	-- ¹
DTX1	.048	1.000	.071	.902	-- ¹	-- ¹	-.571	.063	.428	.173	-- ¹	-- ¹
DTX10A	-- ¹	-- ¹	.357	.266	-- ¹	-- ¹	-.036	1.000	-.178	.621	-- ¹	-- ¹
DTX2	-.393	.216	.142	.710	-- ¹	-- ¹	-.607	.048	.393	.216	-- ¹	-- ¹
DTX3	-.214	.536	-- ¹	-- ¹	-- ¹	-- ¹	-.214	.536	.178	.621	-- ¹	-- ¹
DTX4	.500	.108	.036	1.000	-- ¹	-- ¹	-.286	.386	<.001	1.000	-- ¹	-- ¹
DTX5	-.464	.138	.428	.174	-- ¹	-- ¹	-.750	.013	.071	.902	-- ¹	-- ¹
DTX6	-.250	.458	.464	.138	-- ¹	-- ¹	.071	.902	.214	.536	-- ¹	-- ¹
DTX8A	-.607	.048	-- ¹	-- ¹	-- ¹	-- ¹	-- ¹	-- ¹	.107	.804	-- ¹	-- ¹
Well	Mercury		Molybdenum		Nickel		Selenium		Zinc			
	Tau	p-value	Tau	p-value	Tau	p-value	Tau	p-value	Tau	p-value		
D6	-- ¹	-- ¹	.214	.536	-.107	.804	.464	.138	.071	.902		
D13	-- ¹	-- ¹	.428	.174	.071	.902	-- ¹	-- ¹	-.071	.902		
D17	-- ¹	-- ¹	-.107	.804	<.001	1.000	-.286	.386	-- ¹	-- ¹		
D25	-- ¹	-- ¹	-.250	.458	-.178	.621	.036	1.000	-.250	.458		
D29	.357	.266	-.178	.621	-.464	.138	.464	.138	.107	.804		
D30	-- ¹	-- ¹	<.001	1.000	-.250	.458	.428	.174	-.500	.108		
DTX1	-- ¹	-- ¹	.678	.026	.214	.536	.214	.536	.286	.386		
DTX10A	-- ¹	-- ¹	-.428	.174	-.464	.138	.250	.458	-.357	.266		
DTX2	-- ¹	-- ¹	.286	.386	.143	.710	.464	.138	.071	.902		
DTX3	-- ¹	-- ¹	-.321	.322	.214	.536	.250	.458	.178	.621		
DTX4	-- ¹	-- ¹	<.001	1.000	-.393	.216	.643	.035	-.143	.710		
DTX5	-- ¹	-- ¹	<.001	1.000	-.143	.710	.428	.174	-.036	1.000		
DTX6	-- ¹	-- ¹	-.286	.386	-.321	.322	.250	.458	-.25	.458		
DTX8A	-- ¹	-- ¹	-.250	.458	-.178	.621	-- ¹	-- ¹	-.107	.804		

¹ No concentrations were greater than the laboratory minimum reporting limit.

Table 18. Methods used to analyze streambed-sediment samples collected near Deer Trail, Colorado, 2000

[Samples were analyzed at the National Water Quality Laboratory in Denver, except for constituents footnoted; MRL, minimum reporting level; MDC, minimum detectable concentration (radiochemical samples); ICP, inductively coupled plasma; GFAA, graphite furnace atomic absorption; AA, atomic absorption; ASF, automated segmented-flow spectrophotometry; *, not applicable; mg/kg, milligrams per kilogram; µg/g, micrograms per gram; g/kg, grams per kilogram; pCi/g, picocuries per gram]

Property or Constituent	Units	Analytical method	MRL or MDC
Nutrients			
Nitrogen, ammonia plus organic ¹	mg/kg	Colorimetry, ASF, salicylate-hypochlorite	*
Phosphorus, total as P	mg/kg	Colorimetry, ASF, phosphomolybdate	40
Trace elements			
Aluminum	µg/g	DCP	2.8
Arsenic	µg/g	GFAA	1
Cadmium	µg/g	AA	.1
Chromium	µg/g	AA	.4
Copper	µg/g	AA	2
Lead	µg/g	AA	.1
Mercury	µg/g	AA manual cold vapor	.01
Molybdenum	µg/g	AA	.1
Nickel	µg/g	AA	.1
Selenium	µg/g	AA, Hydride generation, ASF	1
Zinc	µg/g	AA	3.1
Carbons			
Inorganic carbon	g/kg	Coulometric	.2
Organic carbon	g/kg	Wet oxidation	.2
Total carbon	g/kg	Induction furnace (Leco)	.1
Radioactivity			
Gross alpha ²	pCi/g	Thorium-230	*
Gross beta ²	pCi/g	Cesium-137	*
Plutonium-238 ²	pCi/g	Alpha spectrometry	*
Plutonium-239+240 ²	pCi/g	Alpha spectrometry	*

¹ Analyses done by Severn Trent Laboratory in Arvada, Colorado. MRL calculated for each sample.

² Analyses done by Severn Trent Laboratory in Richland, Washington. MDC calculated for each sample.

Table 19. Streambed-sediment data collected near Deer Trail, Colorado, July 17, 2000
 [mg/kg, milligrams per kilogram; µg/g, micrograms per gram; g/kg, grams per kilogram; pCi/g, picocuries per gram; <, less than; ±, plus or minus the analytical uncertainty]

Site	Date	Time	Total nitrogen (mg/kg)	Phos- phorus (mg/kg)	Aluminum (µg/g)	Arsenic (µg/g)	Cadmium (µg/g)	Chromium (µg/g)	Copper (µg/g)	Lead (µg/g)	Mercury (µg/g)	Molyb- denum (µg/g)
Biosolids-applied basin	07/17/00	1500	824	580	8,500	<1	0.11	8.0	12.3	13.1	0.02	0.18
Control basin (no biosolids)	07/17/00	1645	836	540	9,240	<1	.12	10.2	10.5	13.8	.02	.20
Control basin, replicate	07/17/00	1705	730	600	10,400	<1	.14	10.8	12.8	15.4	.02	.21

Site	Date	Time	Nickel (µg/g)	Selenium (µg/g)	Zinc (µg/g)	Carbon, inorganic (g/kg)	Carbon, organic (g/kg)	Gross alpha (pCi/g)	Gross beta (pCi/g)	Plutonium-238 (pCi/g)	Plutonium- 239+240 (pCi/g)
Biosolids-applied basin	07/17/00	1500	10.3	<1	40.8	4.7	7.4	9.8 ± 4.4	44.3 ± 7.7	0 ± 0.002	0.002 ± 0.003
Control basin (no biosolids)	07/17/00	1645	12.5	<1	44.4	4.1	7.2	10.4 ± 4.7	44.2 ± 7.8	0 ± .003	.002 ± .004
Control basin, replicate	07/17/00	1705	12.6	<1	48.0	4.5	8.2	16.9 ± 5.9	44.1 ± 7.6	.001 ± .002	.003 ± .003